

PHASE 1–4 FINAL TESTING CHECKLIST

Agricultural Voice Assistant Backend — Pre-Phase-5 Acceptance Test

Purpose: Systematically validate every implemented layer through Phase 4 before starting the API-based LLM. The Phase 4 report records 69 automated tests passing, but real Gemini authentication/setup and speech transcription were not verified; these external checks are mandatory.

1. Phase 1 — Foundation

■ Full build

Expected: BUILD SUCCESS; 0 failures/errors/skipped. Current report: 69 tests, 0 failures/errors/skipped.

Test: `.\mvnw.cmd --no-transfer-progress clean verify`

■ Application startup

Starts successfully without crashing.

■ Health endpoint

GET /health returns HTTP 200 and {"status":"UP"}.

Test: `Invoke-WebRequest http://localhost:8080/health`

■ Configuration

Server/database settings are environment-driven. No real secret is in application.yaml or .env.example. Application can start without a Gemini key.

2. Phase 2 — Local WebSocket / Exotel Media Layer

■ Connect

ws://localhost:8080/media accepts a connection.

Test: `.\scripts\Test-MediaWebSocket.ps1`

■ connected

Valid connected event is accepted; session remains alive.

■ start

Valid start creates the session and records callSid, streamSid, encoding, sample rate and channels.

■ Duplicate start

Does not create a second/conflicting session.

■ media

Valid Base64 PCM becomes AudioFrame with correct stream/format metadata.

■ Invalid Base64

Rejected safely; server does not crash.

■ Malformed/missing fields

Test missing callSid/streamSid/media/sample rate and malformed JSON; graceful rejection; no raw payload logging.

■ DTMF / mark / clear

Each event is recognized safely according to the implemented protocol behavior.

■ stop

Session state is cleaned up.

■ Duplicate stop

Second stop is harmless/idempotent.

■ Isolation

One failed session cannot affect another session or the application.

3. Phase 3 — Public WSS

■ Start backend

Run application on configured port, normally 8080.

■ Start Quick Tunnel

Create a development-only Cloudflare Quick Tunnel.

Test: `cloudflared tunnel --url http://localhost:8080`

■ Public HTTPS health

<https://trycloudflare.com/health> returns HTTP 200 and status UP.

■ Public WSS

<wss://trycloudflare.com/media> connects.

■ Public lifecycle

connected → start → media → stop works through WSS.

■ TLS/proxy path

Cloudflare TLS terminates and proxies to local <ws://localhost:8080/media>.

■ Cleanup

Stop tunnel; temporary hostname is not treated as permanent.

4. Phase 4 — Audio Normalization

■ 16 kHz PCM16 mono

Passes through without resampling.

■ 8 kHz PCM16 mono

Resamples to 16 kHz; approximately 2× sample count.

■ 24 kHz PCM16 mono

Resamples to 16 kHz; approximately 2/3 sample count.

■ Invalid audio

Reject null/empty, odd byte count, stereo, non-16-bit and unsupported sample rates.

■ Frame size

Oversized input is rejected/bounded.

5. Phase 4 — Chunking & Backpressure

■ 100 ms chunk

$16,000 \text{ samples/s} \times 2 \text{ bytes} \times 0.1 \text{ s} = 3,200 \text{ bytes}$.

■ Partial chunk

Incomplete chunk is retained rather than lost.

■ Final partial chunk

Remaining partial audio is sent before stream-end.

■ Bounded queue

Default capacity 20 chunks (~2 seconds); memory cannot grow without bound.

■ Backpressure

With a slow fake Gemini transport, full queue returns BACKPRESSURE and inbound processing remains non-blocking.

■ No network blocking

WebSocket inbound handling does not wait for Gemini network I/O.

6. Phase 4 — Gemini Protocol & Credentials

■ Fresh key

Revoke any exposed/old credential and create a fresh Gemini key. Never paste it into ChatGPT/Codex or commit it.

■ Local secret

Store only as GEMINI_API_KEY in ignored local .env/deployment secret storage.

■ No key logging

Logs contain no API key, authorization header or authenticated Gemini URI.

■ Model

Configured model is gemini-3.5-transcribe-live.

■ Official endpoint/protocol

Verify the implementation matches Google's current Live Transcription WebSocket protocol; no guessed fields.

■ Setup message

Correct setup config: TEXT response modality and input audio transcription.

■ Audio

Gemini receives PCM16 mono at 16 kHz using the documented audio MIME type.

■ setupComplete — MANDATORY

With a fresh valid key, Gemini must return setupComplete. This is the current Phase 4 blocker.

■ Invalid key

Invalid key fails only the affected STT session; Spring Boot remains alive.

■ Provider disconnect

Gemini disconnect cleans only the affected STT session.

■ Reconnect safety

No retry mechanism can duplicate already-delivered audio/transcripts.

7. Phase 4 — Real Speech Recognition

■ Real Gemini smoke test

Use a valid fresh key. Silence-only testing validates transport, not recognition.

■ Hindi

Say: “■■■■■■ ■■ ■■ ■■ ■■■ ■■■■ ■■?”; verify a sensible Hindi transcript.

■ English

Say: “What is the price of wheat today?”; verify a sensible English transcript.

■ Hinglish

Say: “Mandi mein wheat ka kya bhav chal raha hai?”; verify reasonable code-switched transcription.

■ Hindi + English

Say: “■■■■■■ ■■ market price ■■■■ ■■ ■■■ ■■?”.

■ Agricultural vocabulary

Test wheat, rice, paddy, maize, ■■■■■■, ■■■■, ■■■■, ■■■■, mandi, MSP, quintal, ■■■■■■■■, ■■■■■■, FPO, ■■■■, ■■■■.

■ Interim transcript

Continuous speech produces an INTERIM Transcript.

■ Final transcript

Completed utterance produces a FINAL Transcript.

■ Ordering

Per-session sequence numbers increase monotonically.

■ Privacy

Full transcript text is not logged by default.

8. Phase 4 — Lifecycle, Failure & Concurrency

■ Full lifecycle

START → Gemini setup → media → transcript → STOP → cleanup.

■ Abnormal /media close

Closing the voice socket cleans the Gemini STT session.

■ Gemini failure isolation

One failed Gemini session does not affect other sessions.

■ Multiple sessions

Run 2–3 sessions; audio/transcripts never cross between calls.

■ Duplicate lifecycle

Repeated start/stop events remain safe.

■ Application shutdown

Active STT transports close cleanly on shutdown.

9. Security Review

■ Repository secret scan

Search for real API keys, tokens, passwords and Authorization headers; none may be committed.

- **.env ignored**

Confirm .env is ignored and not tracked.

■ Raw audio

No PCM/Base64 audio in logs.

■ PII

No unnecessary caller/destination phone numbers in logs.

■ Gemini payloads

No complete setup/audio/provider response payloads in logs.

■ Public exposure

Gemini is reached only by the backend; key is never exposed via /media.

■ Failure containment

Malformed provider data or one session failure cannot crash the application.

10. Final End-to-End Acceptance Test

Run only after the preceding checks pass. Use a real speech source.

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Voice → /media WSS → VoiceSession → AudioFrame → PCM normalization → 100 ms queue → Gemini WebSocket →
setupComplete → interim transcript → final transcript → cleanup

```

■ Hindi end-to-end

Say: “■■■■■■■, ■■■■ ■■■■■■ ■■ ■■ ■■ ■■■■ ■■■ ■■■■■■”; verify final transcript.

■ English end-to-end

Say: **"Tell me today's wheat market price."**; verify final transcript.

■ Hinglish end-to-end

Speak a mixed Hindi/English agricultural sentence; verify reasonable transcription.

■ Cleanup

Stop call; VoiceSession and Gemini STT session are both cleaned.

■ Post-call health

/health still returns 200 and application remains operational.

11. Evidence to Record

Area	Evidence
Build	Terminal output: BUILD SUCCESS + test count
Health	HTTP 200 response
Local WS	Test script output
Public WSS	Tunnel + WSS test output
Audio	Normalization/chunk tests
Backpressure	Fake transport result
Gemini	setupComplete with fresh key
Hindi	Final transcript evidence
English	Final transcript evidence
Hinglish	Final transcript evidence
Security	Secret/log scan
Cleanup	Session/resource cleanup evidence

12. GO / NO-GO for Phase 5

- Full Maven build passes.

- /health works.
- Local /media lifecycle works.
- Public WSS lifecycle works.
- Audio normalization passes.
- Chunking and bounded backpressure pass.
- Fresh Gemini key authenticates.
- Gemini setupComplete is observed.
- Real Hindi speech transcribes reasonably.
- Real English speech transcribes reasonably.
- Real Hinglish/code-switching transcribes reasonably.
- Interim and final transcripts work.
- Session cleanup works.
- Security/log review passes.

NO-GO: If setupComplete or real speech transcription fails, fix Phase 4 before starting Phase 5.

Next phase after acceptance: Phase 5 — API-based LLM, with the boundary Transcript → LLMService. Do not add database tools or TTS into Phase 5.